COLLEGE OF ENGINEERING
Central Philippine University
BS Software Engineering Program

SE 3141: SOFTWARE ARCHITECTURE
A.Y. 2026 – 2027

ACTIVITY 4: INTEGRATION ARENA — BUILD, TEST, DEFEND

ACADEMIC INTEGRITY

As a student of Central Philippine University, I pledge to complete this assignment with honesty and integrity, upholding the University's core values of Scientia et Fides — knowledge and faith. I understand that suspected academic misconduct will be reported to the appropriate office and, if established, will result in disciplinary action in accordance with University rules, policies, and procedures. This is a pair activity. The submitted code and diagram must be the original work of the pair, and both members must contribute to and understand the submitted work.

INSTRUCTIONS:

This is a **PAIR ACTIVITY**. Work with **one (1) partner** throughout all three parts. Both members are expected to contribute to the coding, business capability map, and preparation of the defense.

You will build a small, testable Order Service that integrates with two unreliable external systems, map the business capability it supports, and defend both live in a short one-on-one lab session.

There are three parts: **a coding lab (autograded), a business capability map, and a lab defense.**

PART 1 — THE INTEGRATION ARENA (CODING LAB) — 50%

Download the starter kit (mock_services.py, app_template.py, test_lab_autograder.py, run_lab.py). Rename app_template.py to app.py and implement your Order Service level by level, applying the integration patterns from Lesson 1. Run *python run_lab.py* anytime to see your live scoreboard.

Level	Pattern You Implement	Module 4 Concept	Points
1	Call the Payment Gateway and return its result	Synchronous communication, API	30
2	Add idempotency keys and retry-with-backoff on failure	Reliability: idempotency, retries	30
3	Send the confirmation without blocking the response	Asynchronous communication	20
4 (Boss)	Open a circuit breaker after repeated failures and fail fast	Circuit breaker (Module 2 & 4)	20

PART 2 — BUSINESS CAPABILITY MAP — 20%

Using the same system from Part 1 (or the system from Activity 3), create a one-page business capability map (a grid of capabilities grouped by domain, following Figure 2.2 of Module 4). Then write a short paragraph (100–150 words) tracing one business goal down through a capability, a process, and finally the service(s) you built in Part 1 — answering "why does this service exist?" all the way up the chain.

PART 3 — LAB DEFENSE — 30%

Sign up for a 5–7-minute pair defense slot during the lab period. Both members of the pair must be present and participate in the defense.

Bring your working code and your Part 2 diagram. You will run run_lab.py live for the panel and answer questions such as:

- Why is the payment call synchronous but the confirmation asynchronous?
- What would break if you removed the idempotency key check?
- How does your circuit breaker relate to the availability tactics?
- How does your Part 1 service trace back to the business capability in Part 2?

EVALUATION RUBRIC

EVALUATION CRITERION	WEIGHT	EXPECTATIONS
Autograder Score (Part 1)	40%	Levels pass test_lab_autograder.py without modification.
Business Capability Map (Part 2)	15%	Capabilities are correctly grouped by domain; goal-to-service chain is traced clearly.
Defense: Technical Explanation	25%	Can walk through the code and justify each design decision.
Defense: Concept Mastery	20%	Correctly names and applies Module 4 vocabulary when answering questions.

SUBMISSION

- Code: Lastname1_Lastname2_Act4_Code.zip
- Business Architecture: Lastname1_Lastname2_Act4_BizArch.pdf
- Defense: Both members must attend and participate in the same pair defense slot.

Deadline: OCTOBER 6, 2026

Lab Defense Week: OCTOBER 13, 2026